

Robotic Integration Technical Report

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1. Physics & Geometry Analysis

Property	Specification
Volume (mm)3	345862.83
Material Used	steel
Center Of Mass (mm) > X	31.09
Center Of Mass (mm) > Y	50
Center Of Mass (mm) > Z	33.13
Estimated Mass (kg)	2.715
Inertia Tensor (kg) M2 > Ixx	0.004755
Inertia Tensor (kg) M2 > Ixy	0
Inertia Tensor (kg) M2 > Ixz	0.001324
Inertia Tensor (kg) M2 > Iyy	0.004777
Inertia Tensor (kg) M2 > Iyz	0
Inertia Tensor (kg) M2 > Izz	0.00472
Bbox Max	100.11, 100.11, 100.11
Bbox Min	-0.11, -0.11, -0.11
Dimensions (mm) > Width Y	100.22
Dimensions (mm) > Height Z	100.22
Dimensions (mm) > Length X	100.22
Part Diagonal (mm)	173.59
Heuristic Clearance (mm)	260.38

2. Picking Analysis & Suction Strategy

Complexity Score: Low | Valid Suction Faces: 5

Face Index	Area (mm ²)	Normal Vector (x,y,z)
Face 1	8000.00	[1, 0, 0]
Face 2	8000.00	[1, 0, 0]
Face 3	7293.14	[0, 0, 1]
Face 4	7293.14	[0, 0, 1]
Face 5	2000.00	[1, 0, 0]

3. Fanuc Controller Manifest

Manifest Item	Requirement / Standard
Fieldbus Options	R639 - PROFINET I/O Device (Siemens Standard), R510 - EtherNet/IP Adapter (Rockwell Standard), J758 - DeviceNet (Legacy Support)
Software Packages	R632 - PMC (Internal PLC Logic), R663 - iRVision 2D Guide (Vision Tending), R739 - Intelligent Interference Check
Controller Platform	R-30iB Plus (Standard)
Integration Handshake	Deterministic 8-byte Discrete I/O map suggested
Safety Type	heuristic
Required Options > J535	Collision Guard - Required for Tooling Protection
Required Options > J744	Highly Recommended
Required Options > R619	Safe I/O Unit - Required for Hardware Interlocking
Safety Standards	ISO 10218-1:2011, ISO/TS 15066 (Collaborative), ANSI/RIA R15.06
Compliance Rating	Category 3 PLd
Stop Time Estimation Ms	363.6
Heuristic Separation Distance (mm)	709.72
Maintenance Mode	Deterministic Cycle Counting
Battery Replacement	Annual (4x 1.5V D-Size Alkaline)
Recommended Lubricant	Fanuc Vigogrease RE0
Grease Replacement Hrs	11520
Reliability Derating (%)	95

4. Integrator Recommendation: Primary Solution

Robot Specification	Value
Model	CRX-10iA
Series	CRX
Reach (mm)	1249
Ip Rating	IP54
Cycle Data > Feasible	true
Cycle Data > Margin (s)	33
Cycle Data > Estimated (s)	12
Payload (kg)	10
Safety Data > Status	VALID
Safety Data > Severity	NORMAL
Safety Data > Standard	ISO 13855
Safety Data > Safety Type	heuristic
Safety Data > Safety Dist (mm)	709.72
Payload Utilization	47.6

Engineering, Integration & Safety

Integration Parameter	Detail
Gripper Type	parallel
Input Mass (kg)	2.715
Design Reach (mm)	600
Risk Assessment	LOW
Design Payload (kg)	5.24
Input Cog Offset (mm)	67.56
Calculated Torque Nm	1.92
Gripper Allowance (kg)	2.04
Target Plc	GENERIC
Native Protocols	EtherNet/IP (Default)
Required Fanuc Options	None
Safety Dist (mm)	709.72
Iso 10218 Status	VALIDATED
Environmental Rating	IP54
Emergency Stop Category	CAT 3 / PLd
Lubricant	Fanuc Vigogrease RE0
Duty Cycle Risk	Standard

Integration Parameter	Detail
Estimated Mtbh Hours	80000
Grease Interval Hours	7680
Joint Inertia > J4	1.92
Joint Inertia > J5	1.54
Joint Inertia > J6	0.96
Software Manifest	J744 - Dual Check Safety (DCS) J535 - Collision Guard

5. Alternative Robot Configurations

Model	Reach	Payload	Util %	Cycle (s)	Safety Dist
CRX-10iA/L	1418 mm	10 kg	47.6%	12	709.72 mm
M-10iD/10L	1636 mm	10 kg	47.6%	12	709.72 mm
M-10iD/12	1441 mm	12 kg	39.7%	11.61	709.72 mm

6. General Engineering Compliance

Technical Notes	Standard industrial ambient conditions.
Required Ip Rating	IP54
Required Lubricant	Standard Fanuc Grease
Environment Context	general